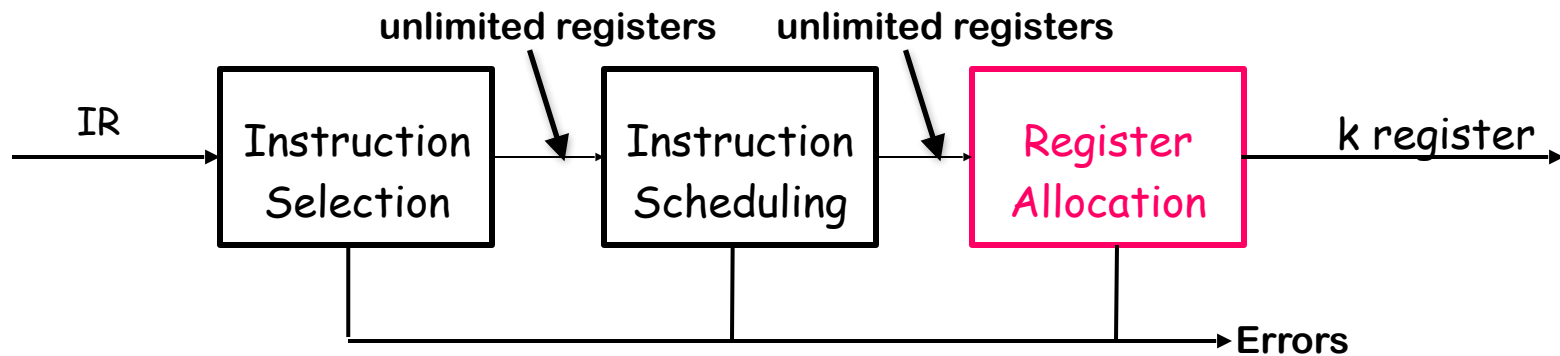


Local Register Allocation

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Register Allocation

Part of the compiler's back end



Spill code: Loads and Stores
inserted by the register allocator

Critical properties

- Produce correct code that uses no more than k registers
- Minimize added work from loads and stores that spill values
- Minimize space used to hold spilled values
- Operate efficiently
 $O(n)$, $O(n \log_2 n)$, maybe $O(n^2)$, but not $O(2^n)$

Notation: The literature on register allocation consistently uses k as the number of registers available on the target system.

Register Allocation

r0 holds base address for local variables

@x is constant offset of x from r0

Consider a fragment of assembly code (or ILOC) for $a - (2 \cdot b)$

```
loadI    2      ⇒ r1    // r1 ← 2
loadAI   r0, @b  ⇒ r2    // r2 ← b
mult     r1, r2  ⇒ r3    // r3 ← 2 · b
loadAI   r0, @a  ⇒ r4    // r4 ← a
sub      r4, r3  ⇒ r5    // r5 ← a - (2 · b)
```

From the allocation perspective, these registers are virtual or pseudo-registers

The Problem

- At each instruction, decide which **values** to keep in
 - Note: each pseudo-register in the example is a value
- Simple if $|\text{values}| \leq |\text{registers}|$
- Harder if $|\text{values}| > |\text{registers}|$
- The compiler must automate this process

Register Allocation

The Task

- At each point in the code, pick the values to keep in registers
- Insert code to move values between registers & memory
 - No transformations (leave that to optimization & scheduling)
- Minimize inserted code — both dynamic & static measures
- Make good use of any extra registers

Allocation versus assignment

- Allocation is deciding which values to keep in registers
- Assignment is choosing specific registers for values
- This distinction is often lost in the literature

The compiler must perform both allocation & assignment

Background issues

The register allocator takes as input a code that is almost completely compiled

It has been scanned, parsed, checked, analysed, optimised, rewritten as target machine code, and, perhaps, scheduled

Many previously made decisions influence the task of the allocator:

- Memory-to-memory versus register-to-register memory model

Additional complexity:

- Allocation vs Assignment
- Register Classes

Register-to -register vs. memory-to-memory

- With a register-to -register earlier phases in the compiler directly encode the knowledge about ambiguous memory references: with this model unambiguous values are kept into virtual registers
- In a register-to -register the code produced by the previous step is not legal
- In a memory-to-memory model, the code is legal before allocation; allocation improve performance
- In a memory-to-memory model the allocator does not have any knowledge and this can limit its ability

Allocation

Allocation is an hard problem that in its general formulation is NP-complete.

The allocation of a single basic block with one size data value can be done in polynomial time under strong hypothesis:

- each value have to be stored to memory at the end of its lifetime (no constant,...)
- the spilling of value has uniform cost

any additional complexity makes the problem NP-complete

Allocation vs. Assignment

- Once we have reduced the demand for registers,
the assignment can be done in polynomial time for a machine with
one kind of registers

Register Classes

- General purpose registers
 - Integer values and memory addresses
 - Floating-point registers (single and double precision)
 - On some architectures also condition code, predicate registers or branch target registers
-
- If the compiler uses different kind of registers for different kinds of data, it can allocate each class independently: the problem can be simplified
 - If the different kinds of data overlap, the compiler must allocate them together: the allocation can become more complex (single and double precision registers for floating-point)

Basic Blocks in Assembly Code (or ILOC)

Definition

- A basic block is a maximal length segment of straight-line (i.e., branch free) code

Importance

- Strongest facts are provable for branch-free code
- If any statement executes, they all execute
- Execution is totally ordered

Ignore, for the moment, exceptions

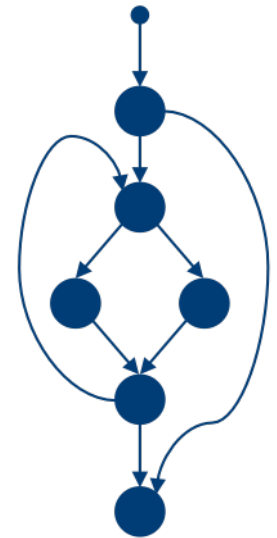


Role of Basic Blocks in Optimization

- Many techniques for improving basic blocks
- Simplest problems
- Strongest methods

Local Register Allocation

- What is "local" ? (different from "regional" or "global")
 - A local transformation operates on basic blocks
 - Many optimizations are done on a local scale or scope
- Does local allocation solve the problem?
 - It produces good register use inside a block
 - Inefficiencies can arise at boundaries between blocks
- How many passes can the allocator make?
 - This is an off-line problem
 - As many passes as it takes, within reason
 - You can do a fine job in a couple of passes



Blocks in a Control-flow Graph (CFG)

Register Allocation

Optimal register allocation is hard

Local Allocation

- Simplified cases $\Rightarrow O(n)$
- Real cases $\Rightarrow$ NP-Complete

Local Assignment

- Single size, no spilling $\Rightarrow O(n)$
- Two sizes $\Rightarrow$ NP-Complete

Global Allocation

- NP-Complete for 1 register
- NP-Complete for k registers
(most sub-problems are NPC, too)

Global Assignment

- NP-Complete

Real compilers face real problems

ILOC

- Pseudo-code for a simple, abstracted RISC machine
 - generated by the instruction selection process
- Simple, compact data structures

$a - 2 \times b$

loadI 2 $\Rightarrow r_1$

loadAI $r_0, @b$ $\Rightarrow r_2$

mult r_1, r_2 $\Rightarrow r_3$

loadAI $r_0, @a$ $\Rightarrow r_4$

sub r_4, r_3 $\Rightarrow r_5$

Nearly assembly code

- simple three-address code
- RISC-like addressing modes
 - I, AI, AO
- unlimited virtual registers
(register-to-register vs memory-to-memory)

ILOC

- Pseudo-code for a simple, abstracted RISC machine
 - generated by the instruction selection process
- Simple, compact data structures

$a - 2 \times b$

loadI	2		r_1
loadAI	r_0	@b	r_2
add	r_1	r_2	r_3
loadAI	r_0	@a	r_4
sub	r_4	r_3	r_5

Quadruples:

- table of $k \times 4$ small integers
- simple record structure
- easy to reorder
- all names are explicit

Observations

The Register Allocator does not need to “understand” the code

- It needs to distinguish definitions from uses
 - Definitions might need to store a spilled value
 - Uses might need to load a spilled value
- ILOC makes definitions and uses pretty clear
 - The assignment arrow, $\Rightarrow$, separates uses from definitions
 - Except on the store operation, which uses all its register operands

store r8 $\Rightarrow$ r1 // MEM(r1) $\leftarrow$ r8

 - That is the point of the arrow!
- Your allocator needs to know, by opcode, how many definitions and how many uses it should see
 - Beyond that, the meaning of the ILOC is somewhat irrelevant to the allocator

Observations

A value is live between its definition and its uses

- Find definitions ($x \leftarrow \dots$) and uses ($y \leftarrow \dots x \dots$)
- From definition to last use is its live range
 - How does a second definition affect this?
- Can represent live range as an interval $[i,j]$ (in block)

Let MAXLIVE be the maximum, over each instruction i in the block, of the number of values (pseudo-registers) live at i .

- If $\text{MAXLIVE} \leq k$, allocation should be easy:
no need to reserve F registers for spilling
- If $\text{MAXLIVE} > k$, some values must be spilled to memory:
need to reserve F registers for spilling

Finding live ranges is harder in the global case

Concrete Example of MAXLIVE

Sample code sequence

loadI	1028	$\Rightarrow$ r1	// r1 $\leftarrow$ 1028	
load	r1	$\Rightarrow$ r2	// r2 $\leftarrow$ MEM(r1)	
mult	r1, r2	$\Rightarrow$ r3	// r3 $\leftarrow$ 1028 $\cdot$ y	
load	x	$\Rightarrow$ r4	// r4 $\leftarrow$ x	
sub	r4, r2	$\Rightarrow$ r5	// r5 $\leftarrow$ x - y	
load	z	$\Rightarrow$ r6	// r6 $\leftarrow$ z	
mult	r5, r6	$\Rightarrow$ r7	// r7 $\leftarrow$ z $\cdot$ (x - y)	
sub	r7, r3	$\Rightarrow$ r8	// r8 $\leftarrow$ z $\cdot$ (x - y) - (1028 $\cdot$ y)	
store	r8	$\Rightarrow$ r1	// MEM(r1) $\leftarrow$ z $\cdot$ (x - y) - (1028 $\cdot$ y)	<u>Store uses this register & defines a memory location.</u>

The code uses 1028 as both an address and as a constant in the computation.

The intent is to create a long live range for pedagogical purposes. Remember, the allocator does not need to understand the computation. It just needs to preserve the computation.

Concrete Example of MAXLIVE

Live ranges in the example

loadI	1028	⇒ r1	// r1		
load	r1	⇒ r2	// r1 r2		
mult	r1, r2	⇒ r3	// r1 r2 r3		
load	x	⇒ r4	// r1 r2 r3 r4		
sub	r4, r2	⇒ r5	// r1 r3 r5		
load	z	⇒ r6	// r1 r3 r5 r6		
mult	r5, r6	⇒ r7	// r1 r3 r7		
sub	r7, r3	⇒ r8	// r1 r8		
store	r8	⇒ r1	//		

Remember, r1 is a use,
not a definition

A pseudo-register is live
after an operation if it has
been defined & has a use in
the future

Concrete Example of MAXLIVE

Live ranges in the example

loadI	1028	⇒ r1	// r1	
load	r1	⇒ r2	// r1 r2	
mult	r1, r2	⇒ r3	// r1 r2 r3	
load	x	⇒ r4	// r1 r2 r3 r4	
sub	r4, r2	⇒ r5	// r1 r3 r5	
load	z	⇒ r6	// r1 r3 r5 r6	
mult	r5, r6	⇒ r7	// r1 r3 r7	
sub	r7, r3	⇒ r8	// r1	r8
store	r8	⇒ r1	//	

MAXLIVE is 4

Remember, r1 is a use, not a definition

Compute these "live" sets in a backward pass over the code.

Start with live as the empty set.

At each op, remove target & add operands

Local Allocation: Top-down Versus Bottom-up

Top-down allocator

- Work from external notion of what is important
- Assign registers in priority order
- Save some registers for the values relegated to memory

Bottom-up allocator

- Work from detailed knowledge about problem instance
- Incorporate knowledge of partial solution at each step
- Handle all values uniformly

Top-down Allocator

The idea

- The most heavily used values should reside in a register
- Reserve registers for use in spills, say r registers

Algorithm

- Count the number of occurrences of each virtual register in the block (from 2 to $\text{maxlength}(\text{block})$)
 - Sort the registers according to the previous info
 - Allocate first $k - r$ values to registers
 - Rewrite code to reflect these choices
- Move values with no register into memory
(add LOADs & STOREs)

Programmers applied this idea by hand in the 70's & early 80's

Top-down Allocator

How many registers must the allocator reserve?

- Need registers to compute spill addresses & load values
- Number depends on target architecture
 - Typically, must be able to load 2 values
- Reserve these registers for spilling

What if $k - r < |\text{values}| < k$?

- Remember that the underlying problem is NP-Complete
- The allocator can either
 - Check for this situation
 - Adopt a more complex strategy
 - Accept the fact that the technique is an approximation

Back to the Example

Top down (3 registers)

r1 is used more
often than r3

loadI	1028	⇒ r1	// r1		
load	r1	⇒ r2	// r1 r2		
mult	r1, r2	⇒ r3	// r1 r2 r3		
load	x	⇒ r4	// r1 r2 r3 r4		
sub	r4, r2	⇒ r5	// r1 r3 r5		
load	z	⇒ r6	// r1 r3 r5 r6		
mult	r5, r6	⇒ r7	// r1 r3 r7		
sub	r7, r3	⇒ r8	// r1 r8		
store	r8	⇒ r1	//		

Note that this assumes that no extra register is needed for spilling

Back to the Example

Top down (3 registers, need 2 for operands)

r1 is used more
often than r3

loadI	1028	⇒ r1	// r1
load	r1	⇒ r2	// r1 r2
mult	r1, r2	⇒ r3	// r1 r2 r3

load	x	⇒ r4	// r1 r2 r3 r4
sub	r4, r2	⇒ r5	// r1 r3 r5
load	z	⇒ r6	// r1 r3 r5 r6
mult	r5, r6	⇒ r7	// r1 r3 r7

spill r3

sub	r7, r3	⇒ r8	// r1
store	r8	⇒ r1	//

r8 restore r3

Note that this assumes that no extra register is needed for spilling

An Example

Top down (3 registers, need 2 for operands)

loadI	1028	⇒ r1	// r1	
load	r1	⇒ r2	// r1 r2	
mult	r1, r2	⇒ r3	// r1 r2 r3	
store	r3	⇒ 16	// r1 r2	
load	x	⇒ r4	// r1 r2	r4
sub	r4, r2	⇒ r5	// r1	r5
load	z	⇒ r6	// r1	r5 r6
mult	r5, r6	⇒ r7	// r1	r7
load	16	⇒ r3	// r1	r7
sub	r7, r3	⇒ r8	// r1	r8
store	r8	⇒ r1	//	

r3 becomes two minimal live ranges ...

"spill" and "restore" become stores and loads

An Example

Top down (3 registers)

```
loadI 1028 ⇒ r1
load  r1    ⇒ r2
mult  r1, r2 ⇒ r3
store r3    ⇒ 16
load  x     ⇒ r4
sub   r4, r2 ⇒ r5
load  z     ⇒ r6
mult  r5, r6 ⇒ r7
load  16    ⇒ r3
sub   r7, r3 ⇒ r8
store r8    ⇒ r1
```

// r1

// r1 r2

// r1 r2

// r1 r2

// r1 r2

// r1

// r1

// r1

// r1

// r1

//

r3

r4

r5

r5 r6

r7

r7

r8

r3 becomes two
minimal live ranges ...

At most 3
values live at
each point

The two short versions of r3 each overlap with fewer values, which simplifies the allocation problem. Such "spilling" will (eventually) create a code where the allocator can succeed.

An Example

Top down (3 registers)

loadI	1028	⇒ r1	// r1		
load	r1	⇒ r2	// r1 r2		
mult	r1, r2	⇒ r3	// r1 r2	r3	
<u>store</u>	<u>r3</u>	<u>⇒ 16</u>	// r1 r2		
loadI	x	⇒ r4	// r1 r2	r4	
sub	r4, r2	⇒ r5	// r1		r5
loadI	z	⇒ r6	// r1		r5 r6
mult	r5, r6	⇒ r7	// r1		r7
<u>load</u>	<u>16 ⇒ r3</u>	// r1	<u>r3</u>	r7	<u>possible delay</u>
sub	r7, r3	⇒ r8	// r1		r8
store	r8	⇒ r1	//		

This code is slower than the original, but it works correctly on a target machine with only three (available) registers.

Correctness is a virtue.

Weakness of the top down approach to allocation

- A physical register is dedicated to a virtual register for an entire block

Bottom-up Allocator

The idea:

- Focus on replacement rather than allocation
- Keep values used "soon" in registers

Algorithm (not optimal!):

- Start with empty register set
- Load on demand
- When no register is available, free one

Replacement:

- Spill the value whose next use is farthest in the future
- Prefer clean values (not to be stored that are constant or values already in memory) to dirty values (that need to be stored).

An Example

Bottom up (3 registers)

loadI	1028	⇒ r1	// r1			
load	r1	⇒ r2	// r1 r2			
mult	r1, r2	⇒ r3	// r1 r2 r3			
loadI	x	⇒ r4	// r1 r2 r3 r4			
sub	r4, r2	⇒ r5	// r1 r3 r5			
loadI	z	⇒ r6	// r1 r3 r5 r6			
mult	r5, r6	⇒ r7	// r1 r3 r7			
sub	r7, r3	⇒ r8	// r1 r8			
store	r8	⇒ r1	//			

All registers are used
at this point



Note that this assumes that no extra register is needed for spilling

An Example

Bottom up (3 registers; need 2 for operands)

loadI	1028	⇒ r1	// r1			
load	r1	⇒ r2	// r1 r2			
mult	r1, r2	⇒ r3	// r1 r2 r3			
loadI	x	⇒ r4	// r1 r2 r3 r4			
sub	r4, r2	⇒ r5	// r1 r3 r5		store r1	
loadI	z	⇒ r6	// r1 r3 r5 r6			
mult	r5, r6	⇒ r7	// r1 r3 r7			
sub	r7, r3	⇒ r8	// r1 r8			
store	r8	⇒ r1	//		restore r1	

Note that this assumes that no extra register is needed for spilling

An Example

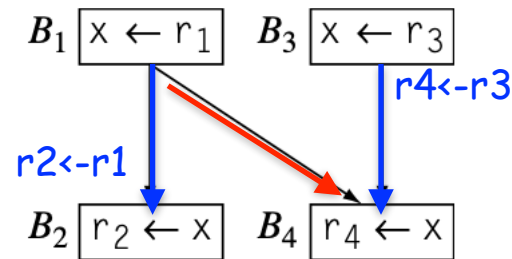
Bottom up (3 registers; need 2 for operands)

loadI	1028	⇒ r1	// r1	
load	r1	⇒ r2	// r1 r2	
mult	r1, r2	⇒ r3	// r1 r2 r3	
store	<u>r1</u>	⇒ <u>20</u>	// r2 r3	
loadI	x	⇒ r4	// r2 r3 r4	
sub	r4, r2	⇒ r5	// r3 r5	At most 3 values <u>live</u> at each point
loadI	z	⇒ r6	// r3 r5 r6	
mult	r5, r6	⇒ r7	// r3 r7	
sub	r7, r3	⇒ r8	// r8	
load	<u>20</u>	⇒ <u>r1</u>	// r1	
store	r8	⇒ r1	//	

The two short versions of r1 each overlap with fewer values, which simplifies the allocation problem. Such "spilling" will (eventually) create a code where the allocator can succeed.

From local algorithms to regional algorithms

- Extending local algorithms to regional ones can be difficult



- the only solution is to store back in memory the value of x at the end of B_1 and B_2
- while we could add register to register operation for blue arrow no possibility exists for the red arrow